Given,

Min. Support = 2
Min. Confidence = 0.7

Iteration-1

Create a table containing support count of each item present in dataset called C_1 (Candidate Set)

C_1

~~Mean~~ Items	Support Count
M_1	6
M_2	7
M_3	6
M_4	2
M_5	2

Since, Min. Support = 2. In L_1, write the items which have support count ≥ 2

L_1

Items	Support Count
M_1	6
M_2	7
M_3	6
M_4	2
M_5	2

Iteration-2

Generate Candidate Set C_2 using L_1 (Join step)

C_2

Items	Support Count
M_1, M_2	4
M_1, M_3	4
M_1, M_4	1
M_1, M_5	2
M_2, M_3	4

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M_2, M_4	2
M_2, M_5	2
M_3, M_4	0
M_3, M_5	1
M_4, M_5	0

L_2

Items	Support count
M_1, M_2	4
M_1, M_3	4
M_1, M_5	2
M_2, M_3	4
M_2, M_4	2
M_2, M_5	2

C_3

Items	Support count
M_1, M_2, M_3	2
M_1, M_2, M_5	2
M_1, M_2, M_4	1
M_1, M_3, M_5	1
M_2, M_3, M_4	1
M_2, M_3, M_5	1
M_2, M_4, M_5	0

L_3

Items	Support count
M_1, M_2, M_3	2
M_1, M_2, M_5	2

C_4

Items	Support count
M_1, M_2, M_3, M_5	1 which is not ≥ 2

So, we take L_3 as final

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Finally

Txn	Support
M_1, M_2, M_3	2
M_1, M_2, M_5	2

Taking $\{M_1, M_2, M_3\}$

Association Rule	Support	Confidence
$M_1 \wedge M_2 \rightarrow M_3$	2	$2/4 = 0.5$
$M_1 \wedge M_3 \rightarrow M_2$	2	$2/4 = 0.5$
$M_2 \wedge M_3 \rightarrow M_1$	2	$2/4 = 0.5$
$M_1 \rightarrow M_2 \wedge M_3$	2	$2/6 = 0.33$
$M_2 \rightarrow M_1 \wedge M_3$	2	$2/7 = 0.28$
$M_3 \rightarrow M_1 \wedge M_2$	2	$2/6 = 0.33$

Taking $\{M_1, M_2, M_5\}$

$M_1 \wedge M_2 \rightarrow M_5$	2	$2/4 = 0.5$
$M_1 \wedge M_5 \rightarrow M_2$	2	$2/2 = 1$ ✓
$M_2 \wedge M_5 \rightarrow M_1$	2	$2/2 = 1$ ✓
$M_1 \rightarrow M_2 \wedge M_5$	2	$2/6 = 0.33$
$M_2 \rightarrow M_1 \wedge M_5$	2	$2/7 = 0.28$
$M_5 \rightarrow M_1 \wedge M_2$	2	$2/2 = 1$ ✓

Since, Min. Confidence = 0.7

Final Rule:

$M_1 \wedge M_5 \rightarrow M_2$

$M_2 \wedge M_5 \rightarrow M_1$

$M_5 \rightarrow M_1 \wedge M_2$

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